GCC Database Partner API

Partner Integration Guide

Version: 1.1 **Date:** [MONTH YEAR]

Prepared for: [PARTNER NAME] Integration Team **Prepared by:** GreenCircle Certified

CONFIDENTIAL — For authorized [PARTNER NAME] integration personnel only

INTERNAL — REMOVE BEFORE SENDING This is the generic partner template. Before issuing to a partner: (1) fill every [HIGHLIGHTED PLACEHOLDER], (2) resolve or delete every red TBD block, (3) confirm the §4 validity model suits this partner's use case — badging works with it, claim validation may not. §8 supplier/parent-company is settled: none is held, so none is sent.

Settled since the last revision: the production host is `database.greencirclecertified.com` — a dedicated subdomain, because the application cannot be hosted on the same server as the WordPress site. The former `greencirclecertified.com/database` path is retired and was never activated. The base URL is the same for every partner, so it is no longer a placeholder; only the key, rate limit, scope and expiry under *Your Connection* vary per partner.

The rate limit in §2 is settled at 200/hour. **DEF-345 is fixed but not yet deployed** — confirm the fix is live on production before issuing this guide to anyone (see the red block in §2). **Never paste a live API key into this document** — keys are delivered separately by secure channel (see §2).

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Your Connection

These values are specific to the key issued to you. Everything else in this guide describes how the API behaves for every partner; anything that differs between partners is here, and here only.

Setting	Your value
Base URL	`https://database.greencirclecertified.com/api/v1` The GCC certificate database is served from its own subdomain, not from a path under the main `greencirclecertified.com` website. This host is the same for every partner.
Test base URL	`https://gccwebapp.com/api/v1` — [issued on request; see §10 step 3]
API key	`gcc_partner_` + 64-character hash, delivered separately by secure channel — never in this document
Rate limit	[N] requests per hour — see §2 for how to read the limit headers
Certificate status mode	[Presence — the feed contains only currently-certified records] — see §4
Endpoints permitted	[All endpoints in §3] — a key may be scoped to a subset; a path outside your scope returns `403`
Product identifier scope	Identifiers not tagged to any partner, plus those tagged to your account — see §8
Key expiration	[None — or the agreed expiry date]

1. Quick Start

Three steps to get your first data:

1. Get your API key from GreenCircle Certified (format: `gcc_partner_[64-character hash]`).
2. Make your first request:

```
curl -X GET "https://database.greencirclecertified.com/api/v1/certificates/gs1" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

3. You'll receive a JSON response containing the certificates tagged to your partner account — including product identifiers (GTIN, MPN, UPC, EAN, SKU), certification details, and PDF download links.

> **Integration model:** You **pull** data from GCC. GCC does not push to you. Call the API on your own schedule — continuously, nightly, or weekly — to retrieve current certification data. Each call returns the full certificate set visible to your account. Your system is responsible for determining what is new, changed, or removed since your last pull.

2. Authentication

Every request requires the `X-API-Key` header:

```
X-API-Key: gcc_partner_[your key - delivered separately]
```

> **Key handling.** Your API key is a bearer credential — anyone holding it can read your certificate data. Store it in a secrets manager or environment variable, never in source control, ticket text, or email bodies. GCC delivers keys by secure channel only and never embeds them in documentation. If a key is exposed, contact GCC immediately for rotation.

Response Code	Meaning	Action
200	Success	Process the response
401	Invalid or missing API key	Check your key in the `X-API-Key` header
403	Endpoint not permitted	Contact GCC support
404	Resource not found	Verify certificate number or product ID
429	Rate limit exceeded	Wait and retry with exponential backoff
500	Server error	Contact GCC support with request details

Rate Limit

Your hourly request limit is set per key and is stated under Your Connection above. Each request returns up to 100 certificates, so a full pull of the certificate database costs one request per 100 records — budget from that, not from the number of certificates.

Every response carries the current state of your allowance:

Header	Meaning
`X-RateLimit-Limit`	Your ceiling for the current window (200)
`X-RateLimit-Remaining`	Requests left in the current window
`X-RateLimit-Reset`	Unix timestamp when the window resets

The window is a fixed clock hour (UTC), not a rolling window — the allowance resets on the hour rather than 60 minutes after your first call. Exceeding the limit returns `429` with a `Retry-After` header. Read `X-RateLimit-Remaining` and pace accordingly rather than retrying into the limit.

INTERNAL — VERIFY ON PRODUCTION, THEN DELETE **DEF-345 is fixed but not yet deployed.** The limiter previously ran before authentication, so partner keys were throttled as anonymous traffic at 100 requests/hour bucketed by IP address rather than by key — and that is what `X-RateLimit-Limit` returned. The fix was made and verified on localhost:5101 on 2026-08-10, but **has not shipped to production**. Before issuing this guide, call the production API with a partner key and confirm `X-RateLimit-Limit: 200` comes back; until that is verified, the 200/hour figure above is a target rather than observed behaviour. Delete this block once confirmed on production.

3. API Endpoints

Base URL: `https://database.greencirclecertified.com/api/v1`

Every path below is relative to that base URL and is identical in the test environment — only the host differs. Point your client at the correct host and nothing else in your integration changes.

The endpoints below are the full set. **An individual key may be scoped to a subset of them** — a request to a path outside your scope returns `403` rather than `404`, so a 403 on a documented endpoint means a permissions question

for GCC, not a wrong URL. Your scope is stated under Your Connection.

GET `/certificates/gs1` RECOMMENDED

List all certificates in **GS1 JSON-LD** format, following the GS1 US Certification Data Model. Product identifiers are always included; GTIN is promoted to a top-level field and other identifier types appear under `additionalTradeItemId`.
Parameters: `pageNumber` (default 1), `pageSize` (default and maximum 100), `searchTerm`, `sortBy`

GET `/certificates/gs1/{certNumber}`

Get a single certificate by certificate number in GS1 format.
Example: `/certificates/gs1/24-0366`

GET `/certificates`

List all certificates in standard JSON format.
Parameters: `pageNumber`, `pageSize`, `searchTerm`, `sortBy`, `sortDescending`, `includeProductIds` (boolean — set `true` to include product identifiers)

GET `/certificates/number/{certNumber}`

Get a single certificate by certificate number in standard JSON format. Use this to validate a brand-provided certificate or license number against the GCC ledger.

GET `/certificates/search/productid`

Find certificates by product identifier.
Parameters: `productId` (required), `type` (optional — GTIN, UPC, EAN, MPN, SKU, ASIN, ISBN; omit to search all types)
Example: `/certificates/search/productid?productId=00012345678901&type=GTIN`

GET `/certificates/search`

Search certificates by free-text term and/or date ranges.
Parameters: `term`, `effectiveDateFrom`, `effectiveDateTo`, `expirationDateFrom`, `expirationDateTo`, plus pagination parameters

GET `/certificates/expiring`

List certificates expiring within a given window — useful for scheduling revalidation of claims before they lapse.
Parameters: `days` (default 30), plus pagination parameters

`GET` **`/certificates/{certNumber}/download`**

Download the certificate PDF document. Returns `application/pdf`.

4. Response Data Model

Certificate Fields

Field	Type	Description	Example
`certificateNumber`	string	Unique certificate identifier	24-0366
`status`	string	Depends on the status mode under Your Connection. In presence mode — the default — always the literal `"Active"`. See *Certificate Status and Validity* below	Active
`issueDate`	datetime	Certificate effective date	2024-01-15T00:00:00Z
`expirationDate`	datetime	Certificate expiration date	2025-01-15T00:00:00Z
`company`	string	Certified company / brand name	EcoProduct Co
`productName`	string	Certified product name	Energy-Efficient LED Bulb
`certificationType`	string	Certification standard granted	Sustainable Energy Practices
`certificationResults`	string	Result or threshold achieved, where applicable*	32%
`pathway`	string	Certification pathway*	Energy usage reduction
`issuer`	string	Certifying body	GreenCircle Certified
`certificationCountries`	string	Applicable countries**	All
`partner`	string	Your partner association on this certificate	[PARTNER NAME]
`amazonCode`	string	Standardized certification code used by retail channel partners. Present in the payload for all partners; may be blank where not applicable to your program.	greencircle_sustainable_energy_practices
`pdfDownloadUrl`	string	Relative URL to download the certificate PDF	/api/v1/certificates/24-0366/download

* `certificationResults` and `pathway` are not applicable to every certification type and are left blank where not relevant.
** Certificates are not restricted by geography; `certificationCountries` is returned as `All` for partner accounts.

Certificate Status and Validity

The API expresses validity by presence, not by a status value. The `status` field is always the literal string `"Active"` on every certificate you receive. It is not a variable to branch on — it never carries any other value for a partner key.

What varies is *whether the certificate appears at all*. GCC tracks an internal certificate lifecycle, and only the states that mean "currently certified" are exposed to partners. Everything else is filtered out of list responses and returns `404` on single-certificate lookups.

What you receive	What it means
Certificate present, `status: "Active"`	Currently certified. Safe to badge, or to confirm a brand's claim.
Certificate absent from the list / `404` on lookup	Not currently certified — **or** not associated with your partner account.

> **Absence does not tell you why.** A certificate you cannot see may have expired, may be on hold, may be withheld as confidential, may still be in draft, may never have existed, or may simply not be linked to your partner account. These cases are deliberately indistinguishable: collapsing them prevents your key from being used to enumerate certificates outside your scope. If you need to know *why* a specific certificate is not visible, ask GCC — do not infer it from the response.

There is no "revoked" state. GCC's lifecycle has no revocation status, so a revoked certificate can never be reported as revoked. Withdrawal of certification is expressed the only way the contract allows: the certificate stops appearing. Treat disappearance as authoritative.

Expiry. `expirationDate` is returned on every certificate, so you can see an upcoming expiry in advance. But note the asymmetry — once a certificate actually moves to an expired state it leaves the API entirely rather than appearing with an expired status. Do not wait to observe a status change that will never arrive; re-pull on your schedule and treat anything that has dropped out as no longer certified.

INTERNAL — CONFIRM SCOPE, THEN DELETE **This documents current behaviour (active-only), which is option (a) of the original open decision.** Internally: the filter is `CertificateStatusHelper.PublicStatuses` = {*Client Approved - Paid*, *Republished*} applied at `Controllers/Api/v1/CertificatesController.cs:173,201` for lists and `:388,442` for single lookups; the literal `"Active"` is set at `:251,315`. Eight internal states exist and none of them is "revoked."

If this partner's use case is *claim validation* rather than badging, confirm that presence/absence is sufficient for them. Novi's guidance, for example, asks for **Valid / Expired / Revoked-or-invalid** as distinct values — which this contract cannot supply. That is raised as **CR-035** (proposed, not built): it changes the response filter, the frozen `/api/v1/certificates*` shape, and — because no "revoked" state exists — potentially the 8-state workflow itself, so it needs client decisions and Amazon sign-off. Do not promise a partner anything beyond the table above until CR-035 is resolved.

Product Identifier Fields (within `productIds[]`)

Field	Type	Description	Example
`type`	string	GTIN, UPC, EAN, MPN, SKU, ASIN, ISBN	GTIN
`value`	string	The product identifier value	00012345678901
`status`	string	Product identifier status	Active
`partner`	string	Your partner association on this identifier	[PARTNER NAME]
`productName`	string	Product variant name	LED Bulb — 4 Pack

Field	Type	Description	Example
`productSku`	string	Company SKU	LED-4PK-ECO

> **Product-level granularity.** Where GCC holds certification data mapped to individual product variants, those identifiers are returned in `productIds[]`. Where a certificate is not mapped to specific variants, the certificate is still returned with its company, product name, and certification details — functioning as a certificate ledger entry you can validate brand claims against.

5. Request/Response Examples

5.1 List All Certificates (GS1 JSON-LD Format)

Request

```
GET /api/v1/certificates/gs1
X-API-Key: YOUR_API_KEY
```

Response

```
{
  "@context": "https://gs1us-technology.github.io/certification-data-model/cert_context.json",
  "$schema": "https://gs1us-technology.github.io/certification-data-model/cert_schema.json",
  "id": "https://database.greencirclecertified.com/api/v1/certificates/gs1/24-0366",
  "type": "CertificationReport",
  "issueDate": "2024-01-15",
  "issuer": "GreenCircle Certified",
  "products": [
    {
      "id": "https://id.gs1.org/01/00012345678901",
      "type": "Product",
      "gtin": "00012345678901",
      "name": "Energy-Efficient LED Bulb",
      "additionalTradeItemId": [
        { "type": "MPN", "value": "MPN-12345" },
        { "type": "SKU", "value": "LED-BULB-001" }
      ],
      "certificationInfo": {
        "type": "CertificationDetails",
        "agency": "GreenCircle Certified",
        "agencyURL": "https://www.greencirclecertified.com",
        "program": "Sustainable Energy Practices",
        "certificateId": "24-0366",
        "status": "ACTIVE",
        "value": "32%",
        "startDate": "2024-01-15",
        "endDate": "2025-01-15",
        "targetCountries": ["All"],
        "pathway": "Energy usage reduction"
      }
    }
  ]
}
```


> **Reading the two `id` fields.** The report-level `id` is built from the host you called, so the same certificate returns `https://database.greencirclecertified.com/api/v1/…` from production and `https://gccwebapp.com/api/v1/…` from test. The product-level `id` is a canonical `id.gs1.org` URI when the product has a GTIN, and falls back to a GCC-hosted `…/product` URI when it does not. Both are JSON-LD identifiers, not endpoints to fetch — use them to key records, not as links to follow.

5.2 List All Certificates (Standard JSON)

Request

```
GET /api/v1/certificates?includeProductIds=true
X-API-Key: YOUR_API_KEY
```

Response

```
{
  "success": true,
  "pagination": {
    "totalCount": 1,
    "pageNumber": 1,
    "pageSize": 100,
    "totalPages": 1,
    "hasNext": false,
    "hasPrevious": false
  },
  "data": [
    {
      "certificateNumber": "24-0366",
      "status": "Active",
      "issueDate": "2024-01-15T00:00:00Z",
      "expirationDate": "2025-01-15T00:00:00Z",
      "certificationCountries": "All",
      "partner": "[PARTNER NAME]",
      "certificationResults": "32%",
      "pathway": "Energy usage reduction",
      "company": "EcoProduct Co",
      "productName": "Energy-Efficient LED Bulb",
      "certificationType": "Sustainable Energy Practices",
      "amazonCode": "greencircle_sustainable_energy_practices",
      "issuer": "GreenCircle Certified",
      "productIds": [
        {
          "type": "GTIN",
          "value": "00012345678901",
          "status": "Active",
          "partner": "[PARTNER NAME]",
          "productName": "Energy-Efficient LED Bulb",
          "productSku": "LED-BULB-001"
        }
      ],
      "pdfDownloadUrl": "/api/v1/certificates/24-0366/download"
    }
  ]
}
```


5.3 Search by Product Identifier

Request

```
GET /api/v1/certificates/search/productid?productId=00012345678999&type=GTIN
X-API-Key: YOUR_API_KEY
```

Response

```
{
  "success": true,
  "data": [
    {
      "certificateNumber": "24-0512",
      "status": "Active",
      "certificationResults": "60% post-consumer recycled content verified",
      "company": "CleanGreen Industries",
      "productName": "Recycled Paper Towels",
      "certificationType": "Recycled Content",
      "issuer": "GreenCircle Certified",
      "productIds": [
        { "type": "GTIN", "value": "00012345678901", "productName": "Paper Towels - 6 Pack" },
        { "type": "GTIN", "value": "00012345678999", "productName": "Paper Towels - 12 Pack",
          "isMatchedId": true },
        { "type": "MPN",  "value": "RPT-24PK",       "productName": "Paper Towels - 24 Pack" }
      ],
      "pdfDownloadUrl": "/api/v1/certificates/24-0512/download"
    }
  ]
}
```


> Searching for one identifier returns the full certificate with **all** product identifiers visible to your account, not just the one you searched. The `isMatchedId: true` flag marks which identifier matched your query, and the matched identifier is sorted first.

5.4 Validate a Brand-Provided Certificate Number

Request

```
GET /api/v1/certificates/number/24-0366
X-API-Key: YOUR_API_KEY
```

Returns the certificate record if it exists and is visible to your account; returns `404` otherwise.

Reading the result. A `200` confirms the brand's claim: that certificate number is currently certified and associated with your account. A `404` means the claim *cannot be confirmed* — which is not the same as the claim being false. It collapses several distinct cases: expired, on hold, confidential, draft, never issued, mistyped, or valid but not linked to your partner account. See *Certificate Status and Validity* in §4 before deciding what action a `404` should trigger on your side; in particular, a `404` is **not** evidence of revocation.

5.5 Download a Certificate PDF

```
GET /api/v1/certificates/24-0366/download
X-API-Key: YOUR_API_KEY
```

Returns the PDF file with `Content-Type: application/pdf`.

6. Error Handling

Errors return a consistent JSON structure:

```
{
  "success": false,
  "errors": [
    {
      "code": "RESOURCE_NOT_FOUND",
      "message": "Certificate with number 24-9999 not found"
    }
  ]
}
```

Code	Error	Resolution
401	Missing or invalid API key	Ensure the `X-API-Key` header is present with a valid key
403	Insufficient permissions	Your key may not have access to this endpoint
404	Certificate or product ID not found	Verify the certificate number or product identifier exists and is visible to your account
429	Rate limit exceeded	Wait and retry with exponential backoff (see the rate limit in §2)
500	Server error	Contact GCC support with the request details

7. Pagination

The API returns certificate data in pages of up to **100 certificates** per request. Use the `pagination` object in each response to determine whether more pages are available:

```
// Page 1
GET /api/v1/certificates?pageNumber=1&pageSize=100

// Check response
"pagination": {
  "totalCount": 250,
  "pageNumber": 1,
  "pageSize": 100,
  "totalPages": 3,
  "hasNext": true,      // more pages available
  "hasPrevious": false
}

// Page 2
GET /api/v1/certificates?pageNumber=2&pageSize=100
```

Loop until `hasNext` is `false`.

8. Security & Access Scope

Control	Details
Transport	HTTPS/TLS required for all requests
Authentication	API key via `X-API-Key` header
Data Isolation	Your key returns only certificates associated with your partner account. Data belonging to other partners is not accessible, including on certificates shared across multiple partners — co-partner names and their product identifiers are removed from your responses.
Status Filter	Only currently-certified certificates are returned; everything else is filtered out of lists and returns `404` on lookup. `status` is always `"Active"`. See §4.
Rate Limiting	Per key — your limit is stated under Your Connection; mechanics in §2
Key Management	Keys are issued, rotated, and revoked by GCC administrators. Keys may be issued with an expiration date.

> **You will never receive:** other partners' data, internal notes, workflow history, draft certificates, or GCC user information.

Supplier, Manufacturer and Parent-Company Data

None is transmitted, because none is held. GCC records the certified entity as a single flat company name. There is no supplier field, no manufacturer-identity field, and no parent-company or corporate-hierarchy relationship anywhere in the certificate record — so there is nothing of that kind for the API to disclose, and no field-level restriction is required to keep it out.

Specifically:

Field	What it is — and is not
`company`	The name of the entity GCC certified — the brand holding the certification. A single value with no hierarchy above or below it. It is not a manufacturer identity and carries no parent-company reference.
`productIds[].type = "MPN"`	Manufacturer *Part* Number — a product code supplied by the brand, in the same list as ASIN, UPC and GTIN. Despite the name it identifies **a product, not a manufacturer**, and reveals no manufacturing entity.
`productIds[].productSku`	The certified company's own SKU for the item. Its own code, not a supplier's.

Company contact details, address, country and website are recorded by GCC for its own operations and are **never included in any partner response** — they are not part of the contract and no partner endpoint projects them.

> If a partner's terms require a written assurance that manufacturer and parent-company information is not shared onward, this section is that assurance: the data is not in the payload because it is not in the system.

9. Field Reference

Standard JSON vs GS1 JSON-LD Field Mapping

Data	Standard JSON Path	GS1 JSON-LD Path
Certificate Number	`data[].certificateNumber`	`products[].certificationInfo.certificateId`
Status	`data[].status`	`certificationInfo.status`
Issue Date	`data[].issueDate` (ISO 8601 + time)	`issueDate` (date only)
Expiration Date	`data[].expirationDate`	`certificationInfo.endDate`
Certification Results	`data[].certificationResults`	`certificationInfo.value`
Certification Type	`data[].certificationType`	`certificationInfo.program`
Issuer	`data[].issuer`	`issuer` / `certificationInfo.agency`
GTIN	`data[].productIds[].value` (type GTIN/UPC/EAN)	`products[].gtin` (14-digit, zero-padded)
MPN / SKU / other IDs	`data[].productIds[].value`	`products[].additionalTradeItemId[].value`
PDF Link	`data[].pdfDownloadUrl`	(use the standard JSON endpoint)

Product Identifier Types

Type	Format	Example
GTIN	14 digits	00012345678901
UPC	12 digits	012345678901
EAN	13 digits	4006381333931
MPN	Variable	MPN-12345
SKU	Variable	RPT-12PK-ECO
ASIN	10 characters (alphanumeric)	B08XQRT456
ISBN	10 or 13 digits	9780306406157

Available Sort Fields

`sortBy` Value	Description
`certificatenumber`	Sort by certificate number
`issuedate`	Sort by issue date
`expirationdate`	Sort by expiration date
`productname`	Sort by product name
`companyname`	Sort by company name

Add `sortDescending=true` for descending order.

10. Onboarding Checklist

Step	Action	Owner
1	Confirm which certificates are in scope for this partner and tag them to the partner account in the GCC Database	GCC
2	Confirm presence/absence validity (§4) is sufficient for this partner's use case — a claim-validation partner may need Valid/Expired/Revoked as distinct values, which requires a change request. Rate limit is settled at 200/hour, but verify **DEF-345** has shipped or the server will not honour it	GCC
3	Issue a test API key against the GCC test environment, with representative test data	GCC
4	Validate the integration end to end against test data — pull, paginate, parse, download PDFs	Partner
5	Confirm field mapping and any gaps against your internal data model	Partner
6	Issue the production API key by secure channel; revoke the test key	GCC
7	Agree the pull cadence and the support/escalation contact on both sides	Both

> **Test environment.** GCC can provide a separate test key and representative sample data so you can validate your integration before any production data is exchanged. Request this through your GCC contact.

Support

For API key requests, issues, or questions, contact GreenCircle Certified: [GCC support contact]

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